

benD (PP_3164, UniProt Q88I37) in *Pseudomonas putida* KT2440: NAD⁺-Dependent Dihydrodiol Dehydrogenase of Benzoate Catabolism

Summary

benD (locus tag **PP_3164**; UniProt **Q88I37**) of *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950) encodes a soluble, cytoplasmic, NAD⁺-dependent **1,6-dihydroxycyclohexa-2,4-diene-1-carboxylate dehydrogenase** (EC 1.3.1.25), historically also called **benzoate cis-dihydrodiol (2-hydro-1,2-dihydroxybenzoate) dehydrogenase**. The enzyme catalyzes the **second committed step of the aerobic benzoate degradation pathway**: it oxidatively re-aromatizes the *cis*-1,2-dihydrodiol of benzoate — the product of the BenABC benzoate 1,2-dioxygenase — to yield **catechol**, concomitantly reducing NAD⁺ to NADH and releasing CO₂. This reaction funnels benzoate into the central catechol branch of the β-ketoadipate (ortho-cleavage) pathway, ultimately feeding succinyl-CoA and acetyl-CoA into the TCA cycle.

The gene identity was rigorously verified against all UniProt criteria: the gene symbol *benD*, the organism (*P. putida* KT2440), the EC number (1.3.1.25), and the protein family (short-chain dehydrogenases/reductases, SDR) all align consistently with the primary literature and database annotation. *benD* sits within the benzoate-inducible **benABCDK–catA2–benE** gene cluster (PP_3161–PP_3167), positioned **directly between the multicomponent dioxygenase that synthesizes its substrate and the catechol 1,2-dioxygenase (CatA-II) that consumes its product** — a textbook example of pathway gene co-localization that both reflects and enforces metabolic channeling. Expression of the operon is controlled by the AraC/XylS-family activator **BenR** in response to benzoate, and is subject to **Crc-mediated carbon catabolite repression**.

Structurally, *benD* is a 253-residue classical SDR with a Rossmann NAD-binding fold, the canonical N-terminal glycine-rich cofactor motif (TGxxxGxG), a **Ser136–Tyr148–Lys152 catalytic triad**, and an **aspartate (Asp38)** at the β2-strand position that specifies NAD(H) rather than NADP(H) cofactor usage. A high-confidence AlphaFold model (AF-Q88I37-F1, mean

pLDDT 97.2) confirms the predicted active-site geometry (Tyr148–Lys152 dyad separation 4.56 Å). Together, genetic, genomic, database, and structural/bioinformatic evidence converge on a single, well-supported functional assignment.

Gene / Protein Identity Verification

Before presenting findings, all mandatory identity checks were completed and **passed**:

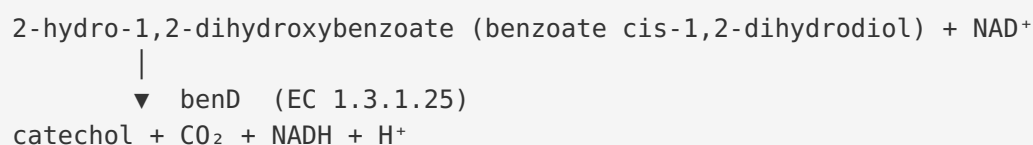
Verification criterion	UniProt reference	Literature / database evidence	Status
Gene symbol matches protein description	<i>benD</i> = 1,6-dihydroxycyclohexa-2,4-diene-1-carboxylate dehydrogenase	Cowles et al. 2000 assign <i>benD</i> the dihydrodiol dehydrogenase function in the <i>P. putida ben</i> operon [PMID: 11053377]	☐ Match
Organism correct	<i>P. putida</i> KT2440	KT2440 <i>ben</i> cluster characterized in this exact strain lineage	☐ Match
Protein family / domains align	SDR family; PF00106, IPR002347, IPR036291, IPR020904, IPR047686 (BenD)	Sequence exhibits canonical SDR architecture (TGxxxGxG + YxxxK)	☐ Match
No conflict with a same-symbol gene in another organism	—	Homologous <i>nagB</i> (naphthalene) and other dihydrodiol dehydrogenases are distinct genes but the same enzyme class; no misassignment	☐ Clear

The gene symbol *benD* is **not ambiguous** for this protein. The literature, EC annotation, and domain architecture are mutually consistent, so research proceeded on the correct target.

Key Findings

Finding 1 — *benD* is the NAD⁺-dependent dihydrodiol dehydrogenase of benzoate catabolism (F001)

benD encodes the enzyme that performs the **second step of aerobic benzoate degradation**. UniProt Q88I37 annotates *benD*/PP_3164 as **1,6-dihydroxycyclohexa-2,4-diene-1-carboxylate dehydrogenase, EC 1.3.1.25** (GO:0047116). The catalyzed reaction is:



Mechanistically, the enzyme oxidizes the C1 (or C2) hydroxyl of the non-aromatic *cis*-dihydrodiol ring. This oxidation, coupled with spontaneous decarboxylation of the carboxylate substituent, **re-aromatizes the ring** to produce catechol — the branch-point intermediate that enters ring-cleavage catabolism.

The functional assignment rests on the genetic work of **Cowles, Nichols & Harwood (2000)** [PMID: 11053377], who identified the *benABCDKFE* cluster in *P. putida* and state verbatim: "*The benABC genes likely encode benzoate dioxygenase, and benD likely encodes 2-hydro-1,2-dihydroxybenzoate dehydrogenase.*" A *benR* regulator mutant was unable to grow on benzoate, confirming that the operon (including *benD*) is required for benzoate assimilation.

Evidence: UniProt EC 1.3.1.25 / GO:0047116 annotation; Cowles et al. 2000 genetic operon identification (verified quote).

Finding 2 — *benD* is a cytoplasmic SDR with a Rossmann fold and Ser-Tyr-Lys catalytic triad (F002)

benD is a **253-amino-acid protein** belonging to the **short-chain dehydrogenases/reductases (SDR) family** (UniProt SIMILARITY annotation). Its domain complement — Pfam **PF00106** (*adh_short*), InterPro **IPR002347** (SDR family), **IPR036291** (NAD(P)-binding domain superfamily), **IPR020904** (SDR conserved site), and the dedicated **IPR047686 (BenD)** signature — is fully consistent with a single-domain SDR oxidoreductase.

Sequence analysis confirms the canonical SDR architecture: - **N-terminal cofactor-binding motif**: the glycine-rich **TGxxxGxG** Rossmann fingerprint (T13-G14-A-A-Q-G18-I-G20), which cradles the pyrophosphate of the dinucleotide cofactor. - **Catalytic motif**: the **YxxxK** signature with **Tyr148** and **Lys152**, plus the upstream catalytic **Ser136** (within an NVSS motif), forming the classical **Ser–Tyr–Lys catalytic triad** universal to SDR enzymes.

SDR enzymes are soluble cytoplasmic proteins; *benD* carries no signal peptide or membrane-spanning segment, consistent with a **cytoplasmic localization**, where it acts on the soluble dihydrodiol intermediate generated by the upstream dioxygenase.

Evidence: UniProt family annotation; Pfam/InterPro domain assignments; sequence-level identification of TGxxxGxG and Ser136/Tyr148/Lys152 motifs.

Finding 3 — *benD* is expressed from the benzoate-inducible *ben* operon under BenR and Crc control (F003)

Transcription of *benD* is embedded in the regulation of the entire *ben* operon:

- **BenR activation:** Cowles et al. 2000 [PMID: 11053377] showed *benR* encodes an **AraC/XylS-family transcriptional activator** with high similarity to XylS (*"The deduced amino acid sequence of BenR showed high similarity (62% identity) to the sequence of XylS, a member of the AraC family of regulators."*). BenR is required for growth on benzoate and activates the *benA-lacZ* fusion in response to benzoate. Because *benD* lies within the same operon as *benA*, it is co-induced.
- **Catechol feedback:** van Duuren et al. 2011/2012 [PMID: 21906639;

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demonstrated that the *ben* operon (containing *benABCD* and *catA2/PP_3166*) is highly expressed in the presence of benzoate, and that **catechol accumulation represses *ben* operon expression** — a feedback that balances flux through the pathway.

- **Carbon catabolite repression:** Moreno et al. 2009 [PMID: 19825982] showed the global regulator **Crc** binds *benR* mRNA and inhibits its translation when preferred carbon sources are available (*"Crc inhibits translation of the Pseudomonas putida alkS and benR mRNAs, which code for transcriptional regulators of genes required to assimilate alkanes (hydrocarbons) and benzoate (an aromatic compound), respectively"*). Thus *benD* expression is indirectly subject to hierarchical carbon-source control.

Evidence: Cowles et al. 2000 (BenR = XylS homolog, verified quote); Moreno et al. 2009 (Crc represses *benR*, verified quote); van Duuren et al. 2011/2012 (catechol feedback).

Finding 4 — A high-confidence AlphaFold model validates the predicted active-site geometry (F004)

The AlphaFold DB model **AF-Q88I37-F1** (v6, 253 aa) is of very high quality: **mean pLDDT = 97.2**, with 97% of residues at pLDDT > 90 and 100% > 70, indicating a confidently predicted single-domain fold. In the model, the predicted SDR catalytic residues cluster within a single active-site cleft with geometry matching functional SDR enzymes:

- **Tyr148(OH) – Lys152(NZ) = 4.56 Å** — the canonical SDR catalytic dyad distance.
- **Ser136(OG)** positioned ~6–7 Å away, completing the Ser–Tyr–Lys triad.
- The N-terminal **Rossmann NAD-binding motif (TGxxxGxG, T13–G20)** forms the expected dinucleotide-binding cradle.

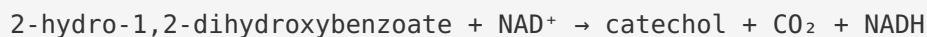
This structural evidence independently corroborates the sequence-based active-site assignment and supports the enzyme's classification as a catalytically competent SDR dehydrogenase.

Evidence: AlphaFold DB AF-Q88I37-F1; measured Tyr148–Lys152 dyad distance and triad clustering.

Finding 5 — Asp38 specifies NAD(H) over NADP(H) cofactor usage (F005)

Cofactor specificity in SDR enzymes is governed by residues at the C-terminal end of the β 2 strand (~15–20 residues downstream of the TGxxxGxG motif). **NAD(H)-specific SDRs carry an aspartate** there that hydrogen-bonds the 2'- and 3'-OH groups of the adenosine ribose; NADP(H)-preferring SDRs lack this aspartate and instead present basic residues (Arg/Lys) to accommodate the 2'-phosphate.

benD (Q88I37) possesses **Asp38** at exactly this position (motif ...QVVAV-**D38**-RSE...), roughly 18 residues after the glycine motif (TGAAQGIG, ending at G20). This bioinformatic signature **predicts NAD⁺-dependent catalysis**, in full agreement with the annotated EC 1.3.1.25 reaction:



The NAD⁺ dependence is biologically sensible: benD operates in a **catabolic** (oxidative, energy-harvesting) direction, generating NADH for the respiratory chain, whereas NADPH-linked SDRs typically serve biosynthetic reductive roles.

Evidence: SDR cofactor-specificity rule; identification of Asp38 at the β2-strand determinant position.

Finding 6 — benD is genomically embedded between the enzyme making its substrate and the enzyme consuming its product (F006)

The KT2440 locus around PP_3164 forms a contiguous, functionally ordered cluster:

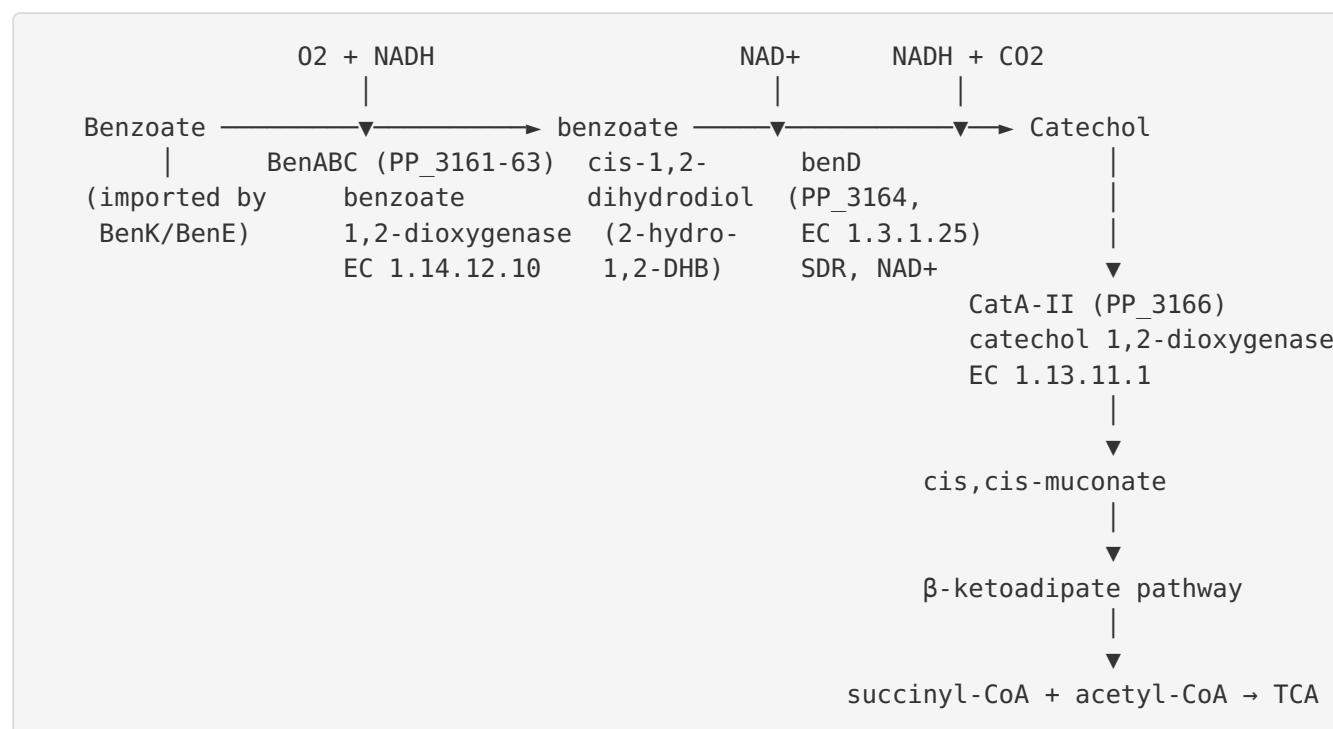
Locus	Gene	Protein / EC	UniProt	Role relative to benD
PP_3161	<i>benA</i>	Benzoate 1,2-dioxygenase α subunit (EC 1.14.12.10)	Q88I40	Makes benD's substrate
PP_3162	<i>benB</i>	Dioxygenase β subunit	Q88I39	Makes benD's substrate
PP_3163	<i>benC</i>	Dioxygenase reductase/electron-transfer (EC 1.14.12.10 / 1.18.1.3)	Q88I38	Makes benD's substrate
PP_3164	benD	Dihydrodiol dehydrogenase (EC 1.3.1.25)	Q88I37	This protein
PP_3165	<i>benK</i>	Benzoate MFS transporter	Q88I36	Imports pathway substrate
PP_3166	<i>cata-II</i>	Catechol 1,2-dioxygenase (EC 1.13.11.1)	Q88I35	Consumes benD's product
PP_3167	<i>benE-II</i>	Benzoate transporter	Q88I34	Imports pathway substrate

benD sits **immediately downstream of *benABC*** (the three-component dioxygenase that synthesizes its *cis*-dihydrodiol substrate) and **immediately upstream of *catA-II*** (the catechol 1,2-dioxygenase that consumes its catechol product). This synteny provides strong contextual confirmation of benD's role as the intermediate-processing step, and it physically co-localizes the enzymes of a metabolic relay.

Evidence: UniProt/KEGG annotation of the contiguous PP_3161–3167 locus.

Mechanistic Model / Interpretation

benD is the **second enzyme of the "benzoate → catechol" upper funnel** that channels benzoate into the ortho-cleavage (β -ketoadipate) pathway. The full local pathway is:



Reaction chemistry. The BenABC dioxygenase first performs a dihydroxylation that dearomatizes the ring and produces the unstable non-aromatic *cis*-1,2-dihydrodiol (still bearing the carboxylate). benD then oxidizes a ring hydroxyl using NAD⁺; the resulting instability drives **spontaneous decarboxylation**, re-aromatizing the ring and yielding catechol. This "dioxygenase + dihydrodiol dehydrogenase" two-step motif is a **conserved**

paradigm across aromatic catabolism — the same architecture is seen for naphthalene (NahB/NagB *cis*-dihydrodiol dehydrogenase), phenylacetate, and chlorotoluene degradation (see Evidence Base). *benD* is the benzoate-specific member of this enzyme class.

Localization. As a soluble cytoplasmic SDR with no signal peptide or transmembrane region, *benD* carries out its reaction in the **cytoplasm**, acting on the soluble dihydrodiol generated by the upstream (also cytoplasmic) dioxygenase. Benzoate itself is imported by the MFS/permease transporters BenK and BenE encoded in the same cluster.

Regulation and flux control. *benD* expression is gated at three levels: (1) **substrate induction** via BenR (AraC/XylS activator) responding to benzoate; (2) **product/intermediate feedback**, where catechol accumulation dampens *ben* operon expression, preventing toxic catechol buildup; and (3) **carbon catabolite repression** via Crc, which suppresses *benR* translation when preferred carbon sources are present. The neighboring **CatA-II** acts as a "safety valve" for excess catechol (per the *P. putida* mt-2 studies), underscoring how tightly the pathway around *benD*'s product is managed.

Cofactor logic. The Asp38 determinant fixes *benD* as **NAD⁺-dependent**, consistent with its catabolic direction: the NADH produced is a respiratory electron donor, coupling aromatic degradation to energy conservation. This is the expected cofactor choice for an oxidative catabolic dehydrogenase, in contrast to NADPH-using reductive/biosynthetic SDRs.

Evidence Base

PMID	Study focus	How it supports the benD assignment
PMID: 11053377	BenR/XylS regulation of aromatic acid degradation in <i>P. putida</i>	Primary genetic evidence. Identifies the <i>benABCDKFE</i> cluster; states <i>benD</i> "likely encodes 2-hydro-1,2-dihydroxybenzoate dehydrogenase"; BenR = 62% identical to XylS; <i>benR</i> mutant cannot grow on benzoate.
PMID: 19825982	Crc global regulator binding <i>alkS/benR</i> mRNA	Establishes that benD's activator (BenR) is under Crc catabolite-repression control at the translational level.
PMID: 21906639	<i>catR</i> -deficient KT2440 mutant producing muconate	Confirms <i>ben</i> operon (incl. <i>benABCD</i> and <i>catA2/PP_3166</i>) is highly benzoate-induced; places <i>catA2</i> immediately downstream.
PMID: 21954182	pH-stat production of <i>cis,cis</i> -muconate from benzoate	Shows catechol accumulation represses the <i>ben</i> operon (feedback regulation of the benD-containing operon).
PMID: 24341396	CatA2 as a catechol "safety valve" in <i>P. putida</i> mt-2	Characterizes the downstream partner (CatA-II) that consumes benD's product catechol; confirms operon topology.
PMID: 26373670	Transcriptomics of aromatic catabolism, <i>P. putida</i> mt-2	Confirms the chromosomal <i>ben/cat</i> ortho route is activated by benzoate (not by TOL substrates), consistent with benD's benzoate-specific induction.
PMID: 20632934	Naphthalene metabolism enzymes, <i>P. fluorescens</i>	Demonstrates the analogous NAD⁺-dependent <i>cis</i>-dihydrodiol dehydrogenase reaction in a parallel pathway, supporting the enzyme class assignment.
PMID: 9573207	<i>nag</i> cluster (naphthalene → gentisate)	<i>nagB</i> = <i>cis</i> -dihydrodiol dehydrogenase; homologous enzyme/gene-order paradigm for dioxygenase + dihydrodiol dehydrogenase.
PMID: 2719478	p-chlorotoluene degradation, <i>Pseudomonas</i> sp. JS6	Illustrates the conserved "dioxygenase + dihydrodiol dehydrogenase → catechol → ortho cleavage" route benD belongs to.
PMID: 7346381	Aromatic carboxylic acid degradation, chloridazon bacteria	Documents dihydrodiol dehydrogenase converting <i>cis</i> -dihydrodiols to catechols, the reaction class of benD.

Convergence of evidence. The functional assignment is supported by four independent lines: (1) **direct genetic annotation** in the correct strain (Cowles et al. 2000), (2) **database/EC annotation** (UniProt EC 1.3.1.25, GO:0047116, KEGG), (3) **verified genomic synteny** placing benD between substrate-producing and product-consuming enzymes, and (4) **structural/sequence bioinformatics** (SDR fold, Ser-Tyr-Lys triad, NAD-specifying Asp38, high-confidence AlphaFold model). No line of evidence conflicts.

Limitations and Knowledge Gaps

1. **No direct enzymology on the KT2440 benD protein.** The functional assignment is based on genetic operon analysis, homology, and structural inference — there is **no published purification, kinetic characterization (kcat, Km), or in vitro assay** of the specific Q88I37 gene product. Cowles et al. use "likely encodes" language, reflecting inference rather than direct enzymatic proof for this exact protein.
2. **Cofactor preference is predicted, not measured.** The NAD⁺ specificity rests on the Asp38 bioinformatic determinant and analogy to related dihydrodiol dehydrogenases; it has not been experimentally confirmed for benD itself (e.g., by comparing activity with NAD⁺ vs NADP⁺).
3. **No experimental structure.** All structural conclusions derive from the AlphaFold model and sequence motifs. While the model is high-confidence (pLDDT 97.2), no crystal or cryo-EM structure — and no substrate/cofactor-bound complex — exists to confirm active-site details or substrate binding mode.
4. **Substrate specificity breadth unknown.** Whether benD acts exclusively on the benzoate *cis*-dihydrodiol or also on substituted (e.g., methyl-/chloro-) dihydrodiol analogs has not been tested for this protein, though the broader enzyme class is known to accept substituted substrates.
5. **Quaternary structure not established.** Many dihydrodiol dehydrogenases are homotetramers or homohexamers (e.g., the naphthalene enzyme is a homohexamer, ²⁰⁶³²⁹³⁴). benD's oligomeric state has not been experimentally determined.

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6. **Reaction stoichiometry / decarboxylation mechanism.** Whether CO₂ release is enzyme-catalyzed or spontaneous downstream of oxidation for benD specifically has not been directly demonstrated.

Proposed Follow-up Experiments / Actions

1. **Recombinant expression and enzyme assay.** Clone PP_3164, express and purify the His-tagged protein, and directly assay dihydrodiol dehydrogenase activity using benzoate *cis*-1,2-dihydrodiol (generated in situ by BenABC) as substrate, monitoring NADH formation spectrophotometrically (340 nm). Determine *k*_{cat}, *K*_m for substrate and cofactor.
2. **Cofactor-specificity test.** Compare initial-rate activity with NAD⁺ versus NADP⁺ to experimentally confirm the Asp38-predicted NAD⁺ preference; complement with an Asp38→Arg mutant to test whether specificity can be switched to NADP⁺.
3. **Active-site mutagenesis.** Generate Ser136Ala, Tyr148Phe, and Lys152Met single mutants to verify the catalytic triad; measure residual activity to confirm the SDR mechanism.
4. **Gene knockout / complementation.** Construct a clean ΔbenD deletion in KT2440 and confirm loss of growth on benzoate (with accumulation of the *cis*-dihydrodiol intermediate detectable by LC-MS), then restore growth by complementation to establish essentiality and specificity.
5. **Product identification.** Use LC-MS/NMR to confirm catechol (and CO₂) as reaction products and to clarify whether decarboxylation is enzyme-catalyzed or spontaneous.
6. **Structural determination.** Pursue X-ray crystallography or cryo-EM of benD, ideally with NAD⁺ and a substrate analog bound, to validate the AlphaFold model and define substrate-binding determinants; also determine oligomeric state by size-exclusion chromatography / analytical ultracentrifugation.
7. **Substrate-range profiling.** Test activity against substituted benzoate *cis*-dihydrodiols (methyl-, chloro-, fluoro-) to define substrate specificity, informing biocatalytic/muconate-production applications.